Category Theory and Functional Programming
Universality and Adjoints

Andrés Sicard-Ramírez

Universidad EAFIT

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Preliminaries

Textbook

Abramsky and Tzevelekos (2011). Introduction to Categories and Categorical Logic.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Adjunctions for Pre-orders

Definition

Let $(P, \preceq_P)$ and $(Q, \preceq_Q)$ be two pre-ordered sets and let f and g be two homomorphisms (monotone maps) forth and back.

$$\begin{tikzcd} P \arrow[r, bend left, "f"] & Q \arrow[l, bend left, "g"] \end{tikzcd}$$

The maps f and g are **adjoint**, denoted $f \dashv g$, when for all $x \in P$, $y \in Q$,

$$f\,x \preceq_Q y \quad \text{iff} \quad x \preceq_P g\,y.$$

The map f is the **left adjoint** and the map g is the **right adjoint** (Awodey and Bauer 2022).

Universal Arrows

Universal Arrows

Definition

Let $G : \mathcal{D} \to \mathcal{C}$ be a functor and let C be an object of $\mathcal{C}$. A **universal arrow** from C to G is a pair (D, η) where D is an object of $\mathcal{D}$ and η is an arrow

$$\eta : C \to G_0\, D$$

Exponentials

Exponentials

Introduction

We shall introduce abstract characterisations of the following facts of set theory:

(i) The functions of a set X to a set Y, denoted Y^X, are also a set.

(ii) A function of two variables $f : X \times Y \to Z$ can be represented as a function of one variable $\lambda f : X \to Z^Y$.

Set theory

Let Z^Y be the set of functions from the set Y to the set Z. For any set X and function $g : X \times Y \to Z$, the following diagrams commutes:

$$
\begin{tikzcd}
Z^Y \\
X \arrow[u, "\Lambda g"]
\end{tikzcd}
\qquad
\begin{tikzcd}
Z^Y \times Y \arrow[r, "\mathrm{eval}_{Y,Z}"] & Z \\
X \times Y \arrow[u, "\Lambda g \times \mathrm{id}_Y"] \arrow[ur, "g"']
\end{tikzcd}
\qquad
\left(g = \mathrm{eval}_{Y,Z} \circ (\Lambda g \times \mathrm{id}_Y)\right)
$$

where

$$\mathrm{eval}_{Y,Z} : (Z^Y \times Y) \to Z := (f, y) \mapsto f\, y \qquad \text{(application)},$$
$$\Lambda g : X \to (Y \to Z) := x\, y \mapsto g(x, y) \qquad \text{(currying)}.$$

Exponentials

Definition

Let $\mathcal{C}$ be a category with binary products, and let Z and Y be objects of $\mathcal{C}$. The pair (Z^Y, eval), where Z^Y is an object of $\mathcal{C}$ and eval is an arrow

$$\mathrm{eval} : (Z^Y \times Y) \to Z$$

is an **exponential object** iff for any object X and arrow $g : X \times Y \to Z$, there is a unique arrow $\Lambda g : X \to Z^Y$ such that the following diagram commutes:

$$\begin{tikzcd}
Z^Y & Z^Y \times Y \arrow[r, "\mathrm{eval}_{Y,Z}"] & Z \\
X \arrow[u, dotted, "\Lambda g"] & X \times Y \arrow[u, dotted, "\Lambda g \times \mathrm{id}_Y"] \arrow[ur, "g"'] &
\end{tikzcd}$$

$$\Big(g = \mathrm{eval}_{Y,Z} \circ (\Lambda g \times \mathrm{id}_Y)\Big)$$

Cartesian Closed Categories

Cartesian Closed Categories

Definition

A category is a **Cartesian closed category** iff it has finite products (i.e. a terminal object and binary products) and exponentials.

References

References

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: 10.1007/978-3-642-12821-9_1 (cit. on p. 2).

Steve Awodey and Andrej Bauer (2022). Introduction to Categorical Logic. Appendix A. Category Theory. Draft version 2022-01-15. URL: https://awodey.github.io/catlog/ (cit. on p. 4).